The Æther-flow Interpretation of Relativity

Substrate-Response / Self-Response Charge-Polarization Candidate
Exact-Preservation Pre-Proto-Subprimitive Stationary-Selection / Cotangent-Connection / Source-Action
Origin Note

Abstract

The exact-preservation proto-subprimitive-substrate / subprimitive-generating shell / lifted connection-cotangent / parallel-transport origin note already pushed the main one-metric charge-polarization line one layer below the recorded subprimitive package. It showed that the recorded proto-subprimitive layer is itself the exact shell-restricted image of a deeper pre-proto-subprimitive layer carrying pre-proto-subprimitive substrate bundles, proto-subprimitive-generating shell sections, lifted subprimitive-generating / primitive-generating / root-generating / pre-ambient-generating / ambient-generating / shell / exact-constraint maps into the same target metrics, pre-proto-subprimitive cotangent-model Liouville one-forms and closed two-form geometry, pre-proto-subprimitive connection one-forms and induced parallel transports, and pre-proto-subprimitive-basic shadow / rate / density / flux / boundary data together with fixed-endpoint pre-proto-subprimitive rate-action geometry. What remained open was the next honest burden below that note: why the pre-proto-subprimitive layer itself should exist at all rather than becoming the new deepest disciplined postulate.

The present note addresses exactly that narrower burden. It keeps fixed the same exact-preservation target: the same $\mathscr{F}_{\mathrm{sel}}$, the same exact-preservation normal form $\mathscr{F}_{\mathrm{deep}}$, the same carrier-origin / primitive-reservoir / microscopic-equilibrium functionals

$$\mathscr{F}_{\mathrm{car}}[\mathfrak{S}, \mathfrak{R}, \mathfrak{F}], \qquad \mathscr{F}_{\mathrm{prim}}[K, J, F], \qquad \mathscr{F}_{\mathrm{mic}}[H, Y],$$

the same Σ^{cmp} solve, the same $\Pi_{\mathrm{free}} H = 0$ outcome, the same observer-equation closure, the same one operative metric, and the same recorded Phase D / E and Phase F gains. Its positive move is not another packaging pass and not stronger promotion language. Instead it introduces a deeper exact-preserving stationary-selection layer:

1. finite-dimensional compact-visible selection bundles with preserved readouts, unit sections, time reversals, and positive metrics;
2. exact-preserving stationary-selection sections whose zero loci are exactly the recorded pre-proto-subprimitive substrate bundles and whose norm-square functionals make those bundles the selected zero-energy stationary states with positive normal gap;
3. lifted proto-subprimitive-generating shell maps, lifted subprimitive-generating shell maps, lifted primitive-generating shell maps, lifted root-generating shell maps, lifted pre-ambient-generating shell maps, lifted ambient-generating shell maps, lifted shell maps, and lifted exact-constraint maps whose pre-proto-subprimitive-shell restrictions are exactly the recorded pre-proto-subprimitive maps;

4. selection cotangent-model Liouville one-forms and closed two-form geometry whose pre-proto-subprimitive-shell restrictions are exactly the recorded pre-proto-subprimitive Liouville and closed two-form geometry;
5. exact-preserving unitary selection connection one-forms whose parallel transports along selection curves restrict exactly to the recorded pre-proto-subprimitive connection one-forms and their induced parallel transports;
6. selection-basic shadow / rate / density / flux / boundary data and fixed-endpoint selection actions whose restrictions are exactly the recorded pre-proto-subprimitive-basic data and the same fixed-endpoint pre-proto-subprimitive rate-action geometry.

At current exact-preservation scope, the pre-proto-subprimitive substrate bundles, their proto-subprimitive-generating shell sections, their lifted subprimitive-generating / primitive-generating / root-generating / pre-ambient-generating / ambient-generating / shell / exact-constraint maps, their pre-proto-subprimitive Liouville / closed-two-form geometry, their pre-proto-subprimitive connection one-forms and induced parallel transports, and their pre-proto-subprimitive-basic shadow / rate / density / flux / boundary data together with their fixed-endpoint pre-proto-subprimitive rate-action geometry are therefore no longer primitive either. Because the present note reproduces the full input layer of the immediately preceding proto-subprimitive note without change, that note applies verbatim. Consequently the same proto-subprimitive layer, the same subprimitive layer, the same primitive layer, the same root layer, the same pre-ambient layer, the same ambient-shell / cotangent-Liouville / connection-holonomy layer, the same process-state and reversible-chain package, the same $\mathscr{F}_{\mathrm{sel}}$, the same exact-preservation normal form, the same carrier-origin / primitive-reservoir / microscopic-equilibrium functionals, the same Σ^{cmp} solve, the same $\Pi_{\mathrm{free}} H = 0$ outcome, the same observer-equation closure, the same one operative metric, and the same recorded Phase D / E and Phase F gains are preserved without change.

The scope remains disciplined. This note does not claim a final microscopic completion, a uniqueness theorem for every deeper realization, or a completed first-principles derivation of GR. Its narrower exact positive claim is only that the recorded pre-proto-subprimitive layer is itself no longer primitive at present scope, but is selected by a still deeper exact-preserving stationary-selection / cotangent-connection / source-action layer while preserving the same downstream package.

1. Introduction

The active one-metric charge-polarization line has now crossed the candidate, benchmark-gate, controlled-limit, residual-sector, redesign, substrate-anchoring, substrate-action, kernel-origin, deeper-selection, primitive-reservoir, carrier-origin, precursor-selection, exact-preservation normal-form, microphysical-Hessian, charge-generator, derivation-algebra, reconfiguration-group, reversible-groupoid, reversible-history, reversible process-window, reversible process-thread, process-state, proto-process, constraint-shell / Liouville-holonomy, ambient-shell / cotangent-connection, pre-ambient-shell / connection-cotangent / parallel-transport, root-shell / lifted connection-cotangent / parallel-transport, primitive-substrate / root-generating shell / lifted connection-cotangent / parallel-transport, subprimitive-substrate / primitive-generating shell / lifted connection-cotangent / parallel-transport, and proto-subprimitive-substrate / subprimitive-generating shell / lifted connection-cotangent / parallel-transport origin steps on the same GR-directed route. The burden therefore sharpens again. It is still not another packaging pass. It is still not stronger promotion language. It is still not, by default, a move back onto the anchored positive-pair side line. It is to go one layer deeper again below the newly recorded pre-proto-subprimitive construction itself and explain why that layer is forced.

That burden has ten parts. First, one must explain why the exact-preserving pre-proto-subprimitive substrate bundles

$$\mathcal{B}^{\mathrm{ppsp}}_{\mathrm{car}}, \qquad \mathcal{B}^{\mathrm{ppsp}}_{\mathrm{cur}}, \qquad \mathcal{B}^{\mathrm{ppsp}}_{\mathrm{hom}}$$

should exist rather than being simply declared. Second, one must explain why the visible pre-proto-subprimitive readout maps, pre-proto-subprimitive unit sections, pre-proto-subprimitive time reversals, preserved positive pre-proto-subprimitive metrics, and compact visible pre-proto-subprimitive fibers should be forced rather than inserted independently. Third, one must explain why the proto-subprimitive-generating shell sections

$$\mathfrak{U}^{\mathrm{ppsp}}_{\mathrm{car}}, \qquad \mathfrak{U}^{\mathrm{ppsp}}_{\mathrm{cur}}, \qquad \mathfrak{U}^{\mathrm{ppsp}}_{\mathrm{hom}}$$

should be forced rather than postulated. Fourth, one must explain why the lifted subprimitive-generating shell maps

$$\widetilde{\mathfrak{T}}^{\mathrm{psp}}_{\mathrm{car}}, \qquad \widetilde{\mathfrak{T}}^{\mathrm{psp}}_{\mathrm{cur}}, \qquad \widetilde{\mathfrak{T}}^{\mathrm{psp}}_{\mathrm{hom}}$$

and the lifted primitive-generating shell maps

$$\widetilde{\mathfrak{G}}^{\mathrm{subprim}}_{\mathrm{car}}, \qquad \widetilde{\mathfrak{G}}^{\mathrm{subprim}}_{\mathrm{cur}}, \qquad \widetilde{\mathfrak{G}}^{\mathrm{subprim}}_{\mathrm{hom}}$$

should themselves be induced rather than separately chosen. Fifth, one must explain why the lifted root-generating shell maps, lifted pre-ambient-generating shell maps, lifted ambient-generating shell maps, lifted shell maps, and lifted exact-constraint maps into the same target metrics should themselves be induced rather than separately chosen. Sixth, one must explain why the pre-proto-subprimitive cotangent-model Liouville one-forms, closed two-form geometry, pre-proto-subprimitive connection one-forms, and induced parallel transports should themselves be induced rather than separately prescribed. Seventh, one must explain why the pre-proto-subprimitive-basic shadow / rate / density / flux / boundary data and the fixed-endpoint pre-proto-subprimitive rate-action geometry should be forced rather than added as independent decorations. Eighth, one must re-show that the same proto-subprimitive layer, the same subprimitive layer, the same primitive layer, the same root layer, the same pre-ambient layer, the same ambient-shell / cotangent-Liouville / connection-holonomy layer, the same process-state and reversible-chain package, the same $\mathscr{F}_{\mathrm{sel}}$, the same exact-preservation normal form, the same carrier-origin / primitive-reservoir / microscopic-equilibrium functionals, the same Σ^{cmp} solve, the same $\Pi_{\mathrm{free}} H = 0$ outcome, the same observer-equation closure, the same one operative metric, and the same recorded Phase D / E and Phase F gains remain intact. Ninth, one must explain why the sharper move is a stationary-selection justification rather than another blind relabeling of a deeper shell. Tenth, one must keep the claim boundary honest while doing so.

The positive claim of the present note is that there is a disciplined way to do exactly that. One introduces a deeper exact-preserving stationary-selection bundle whose visible fibers are compact, whose exact stationary zero shells are precisely the recorded pre-proto-subprimitive substrate bundles, whose lifted proto-subprimitive-generating / subprimitive-generating / primitive-generating / root-generating / pre-ambient-generating / ambient-generating / shell / exact-constraint maps restrict exactly to the recorded pre-proto-subprimitive maps, whose cotangent-model one-forms descend by shell restriction to the recorded pre-proto-subprimitive Liouville geometry, whose skew-Hermitian selection connection one-forms restrict to the recorded pre-proto-subprimitive connection one-forms and generate the recorded pre-proto-subprimitive transports by parallel transport, and whose

selection-basic one-forms / source fields / boundary potentials together with the same fixed-endpoint variational structure restrict to the recorded pre-proto-subprimitive-basic data. The immediately preceding proto-subprimitive note can then be invoked without alteration, so the whole downstream chain is preserved verbatim.

2. Fixed Inputs from the Recorded Completion Line

Definition 1 (Fixed branch and downstream completion target)**.** *The present note keeps fixed the same charge-polarization branch*

$$g^{\mathrm{cand}} = g^{\mathrm{br}} + \Sigma^{\mathrm{IER}} + \Sigma^{\mathrm{resp}}, \qquad g^{\sharp} = g^{\mathrm{br}} + \Sigma^{\mathrm{IER}} + \Sigma^{\mathrm{resp}} + \Sigma^{\mathrm{cmp}}, \tag{1}$$

with lifted variables satisfying

$$\Xi^{*}\widehat{\Sigma}^{\mathrm{IER}} = \Sigma^{\mathrm{IER}}, \qquad \Xi^{*}\widehat{\Sigma}^{\mathrm{resp}} = \Sigma^{\mathrm{resp}}, \qquad \Xi^{*}\widehat{\Sigma}^{\mathrm{cmp}} = \Sigma^{\mathrm{cmp}}. \tag{2}$$

The same completion class and the same free-sector condition are fixed:

$$\widehat{\Sigma}^{\mathrm{cmp}} \in \mathscr{C}_{\mathrm{cmp}}, \qquad \Pi_{\mathrm{free}}\widehat{\Sigma}^{\mathrm{cmp}} = 0. \tag{3}$$

The same substrate and observer completion equations are fixed as well:

$$\widehat{\mathcal{L}}(\widehat{\Sigma}^{\mathrm{cmp}}) = -\widehat{\mathcal{V}}^{\mathrm{res}}, \qquad \mathcal{L}_{g^{\mathrm{br}}}(\Sigma^{\mathrm{cmp}}) = -\mathcal{V}^{\mathrm{res}}_{\mu\nu}. \tag{4}$$

Every statement below must preserve these equations together with the same one operative metric $g^{\sharp}$.

Definition 2 (Recorded pre-proto-subprimitive geometry)**.** *The immediately preceding proto-subprimitive-substrate / subprimitive-generating shell / lifted connection-cotangent / parallel-transport origin note fixes the following exact-preserving input layer for every sector label $\sigma \in \mathsf{S} = \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$.*

1. *Finite-dimensional exact-preserving pre-proto-subprimitive substrate bundles*

$$\mathcal{B}^{\mathrm{ppsp}}_{\sigma}$$

with visible pre-proto-subprimitive readout maps

$$\Pi^{\mathrm{ppsp}}_{\sigma} : \mathcal{B}^{\mathrm{ppsp}}_{\sigma} \to \mathcal{O}_{\sigma},$$

exact-preserving pre-proto-subprimitive unit sections

$$u^{\mathrm{ppsp}}_{\sigma} : \mathcal{O}_{\sigma} \to \mathcal{B}^{\mathrm{ppsp}}_{\sigma},$$

involutive exact-preserving pre-proto-subprimitive time reversals

$$\chi^{\mathrm{ppsp}}_{\sigma} : \mathcal{B}^{\mathrm{ppsp}}_{\sigma} \to \mathcal{B}^{\mathrm{ppsp}}_{\sigma},$$

and preserved positive pre-proto-subprimitive metrics

$$B^{\mathrm{ppsp}}_{\sigma}.$$

Each visible pre-proto-subprimitive fiber $(\Pi_\sigma^{\mathrm{ppsp}})^{-1}(x)$ *is compact.*

2. *Finite-dimensional real pre-proto-subprimitive target spaces*

$$\mathcal{X}_\sigma^{\mathrm{ppsp}}$$

with positive metrics

$$x_\sigma^{\mathrm{ppsp}},$$

and exact-preserving proto-subprimitive-generating shell sections

$$\mathfrak{U}_\sigma^{\mathrm{ppsp}} : \mathcal{B}_\sigma^{\mathrm{ppsp}} \to \mathcal{X}_\sigma^{\mathrm{ppsp}}$$

whose zero loci are exactly the current proto-subprimitive substrate bundles $\mathcal{N}_\sigma^{\mathrm{psp}}$*. These sections carry the same recorded positive pre-proto-subprimitive gap bounds* $\omega_\sigma^{\mathrm{ppsp}} > 0$ *on the current proto-subprimitive shells.*

3. *Exact-preserving lifted subprimitive-generating shell maps*

$$\widetilde{\mathfrak{T}}_\sigma^{\mathrm{psp}} : \mathcal{B}_\sigma^{\mathrm{ppsp}} \to \mathcal{T}_\sigma^{\mathrm{psp}}$$

into the same positive target spaces with the same positive target metrics t_σ^{psp}*. Their restrictions to the current proto-subprimitive shells are exactly the recorded subprimitive-generating shell sections* $\mathfrak{T}_\sigma^{\mathrm{psp}}$ *and carry the same recorded positive proto-subprimitive gap bounds* $\nu_\sigma^{\mathrm{psp}} > 0$ *on the current subprimitive shells.*

4. *Exact-preserving lifted primitive-generating shell maps*

$$\widetilde{\mathfrak{G}}_\sigma^{\mathrm{subprim}} : \mathcal{B}_\sigma^{\mathrm{ppsp}} \to \mathcal{V}_\sigma^{\mathrm{subprim}}$$

into the same positive target spaces with the same positive target metrics $v_\sigma^{\mathrm{subprim}}$*. Their restrictions to the current proto-subprimitive shells are exactly the recorded lifted primitive-generating shell maps* $\check{\mathfrak{G}}_\sigma^{\mathrm{subprim}}$ *and carry the same recorded positive subprimitive gap bounds* $\mu_\sigma^{\mathrm{subprim}} > 0$ *on the current primitive shells.*

5. *Exact-preserving lifted root-generating shell maps*

$$\widetilde{\mathfrak{P}}_\sigma^{\mathrm{prim}} : \mathcal{B}_\sigma^{\mathrm{ppsp}} \to \mathcal{W}_\sigma^{\mathrm{prim}}$$

into the same positive target spaces with the same positive target metrics q_σ^{prim}*. Their restrictions to the current proto-subprimitive shells are exactly the recorded lifted root-generating shell maps* $\check{\mathfrak{P}}_\sigma^{\mathrm{prim}}$ *and carry the same recorded positive primitive gap bounds* $\lambda_\sigma^{\mathrm{prim}} > 0$ *on the current root shells.*

6. *Exact-preserving lifted pre-ambient-generating shell maps*

$$\widetilde{\mathfrak{Z}}_\sigma^{\mathrm{root}} : \mathcal{B}_\sigma^{\mathrm{ppsp}} \to \mathcal{Z}_\sigma^{\mathrm{root}}$$

into the same positive target spaces with the same positive target metrics p_σ^{root}*. Their restrictions to the current proto-subprimitive shells are exactly the recorded lifted pre-ambient-generating*

shell maps $\check{\mathfrak{Z}}_\sigma^{\mathrm{root}}$ *and carry the same recorded positive root gap bounds* $\kappa_\sigma^{\mathrm{root}} > 0$ *on the current pre-ambient shells.*

7. *Exact-preserving lifted ambient-generating shell maps*

$$\widetilde{\mathfrak{S}}_\sigma^{\mathrm{orig}} : \mathcal{B}_\sigma^{\mathrm{ppsp}} \to \mathcal{H}_\sigma^{\mathrm{orig}}$$

into the same positive target spaces with the same positive target metrics n_σ^{orig}*. Their restrictions to the current proto-subprimitive shells are exactly the recorded lifted ambient-generating shell maps* $\check{\mathfrak{S}}_\sigma^{\mathrm{orig}}$ *and carry the same recorded positive ambient-generating gap bounds* $\epsilon_\sigma^{\mathrm{orig}} > 0$ *on the current ambient shells.*

8. *Exact-preserving lifted shell maps*

$$\widetilde{\mathfrak{D}}_\sigma^{\mathrm{root}} : \mathcal{B}_\sigma^{\mathrm{ppsp}} \to \mathcal{E}_\sigma$$

into the same positive shell target spaces with the same positive shell target metrics m_σ^{sh}*. Their restrictions to the current proto-subprimitive exact-constraint shells are exactly the recorded lifted shell maps* $\check{\mathfrak{D}}_\sigma^{\mathrm{root}}$ *and carry the same recorded positive shell-gap bounds* $\eta_\sigma > 0$ *on the current seed shells.*

9. *Exact-preserving lifted exact-constraint maps*

$$\widetilde{\mathfrak{C}}_\sigma^{\mathrm{root}} : \mathcal{B}_\sigma^{\mathrm{ppsp}} \to \mathcal{K}_\sigma$$

into the same positive target spaces with the same positive target metrics m_σ^{con}*. Their restrictions to the current proto-subprimitive shells are exactly the recorded lifted exact-constraint maps* $\check{\mathfrak{C}}_\sigma^{\mathrm{root}}$ *and carry the same recorded positive exact-constraint gap bounds* $\delta_\sigma > 0$*.*

10. *Exact-preserving cotangent realizations*

$$\widetilde{\Psi}_\sigma^{\mathrm{ppsp}} : \mathcal{B}_\sigma^{\mathrm{ppsp}} \to T^*\mathcal{Q}_\sigma$$

producing the pre-proto-subprimitive Liouville one-forms

$$\boldsymbol{\Lambda}_\sigma^{\mathrm{ppsp}}$$

and the closed pre-proto-subprimitive two-forms

$$\boldsymbol{\Xi}_\sigma^{\mathrm{ppsp}}.$$

Their proto-subprimitive-shell restrictions are exactly the recorded proto-subprimitive Liouville one-forms $\boldsymbol{\Lambda}_\sigma^{\mathrm{psp}}$ *and proto-subprimitive closed two-form geometry* $\boldsymbol{\Xi}_\sigma^{\mathrm{psp}}$*.*

11. *Exact-preserving skew-Hermitian pre-proto-subprimitive connection one-forms*

$$\mathcal{F}_\sigma^{\mathrm{conn}} \in \Omega^1(\mathcal{B}_\sigma^{\mathrm{ppsp}}; \mathfrak{u}(\mathscr{U}_\sigma))$$

and the induced exact-preserving unitary parallel transports

$$\mathbb{X}_\sigma$$

attached to piecewise C^1 pre-proto-subprimitive curves, identity on constant curves, multiplicative under concatenation, inverse under the corresponding pre-proto-subprimitive time reversals, and exact-preserving on the fixed branch. Their proto-subprimitive-shell restrictions are exactly the recorded proto-subprimitive connection one-forms $\mathcal{E}_\sigma^{\mathrm{conn}}$ and induced proto-subprimitive parallel transports $\mathbb{W}_\sigma$.

These data are exact fixed inputs for the present note. The only admissible positive result is to derive or justify why this full layer is itself forced while keeping it unchanged.

Definition 3 (Recorded pre-proto-subprimitive basic source and fixed-endpoint action data)**.** *Assume the recorded pre-proto-subprimitive geometry of Definition 2 is fixed. For each $\sigma \in \mathsf{S}$, the same recorded note fixes:*

1. *exact-preserving pre-proto-subprimitive shadow one-forms*

$$\varsigma_\sigma^{\mathrm{ppsp}} \in \Omega^1(\mathcal{B}_\sigma^{\mathrm{ppsp}}; \mathcal{Y}_\sigma^{\mathrm{st}}),$$

invariant under the corresponding pre-proto-subprimitive time reversals, annihilating vectors normal to the proto-subprimitive-generating shell locus $\mathcal{N}_\sigma^{\mathrm{psp}}$, annihilating vertical vectors tangent to the visible pre-proto-subprimitive fibers, and preserved along the current proto-subprimitive characteristic curves singled out on the primitive exact-constraint shells by the restricted proto-subprimitive Liouville / exact-constraint geometry;

2. *exact-preserving pre-proto-subprimitive rate one-forms*

$$\boldsymbol{\lambda}_\sigma^{\mathrm{ppsp}} \in \Omega^1(\mathcal{B}_\sigma^{\mathrm{ppsp}}),$$

with the same invariance and annihilation properties;

3. *exact-preserving pre-proto-subprimitive density and pre-proto-subprimitive flux source fields*

$$\boldsymbol{q}_\sigma^{\mathrm{ppsp}} : \mathcal{B}_\sigma^{\mathrm{ppsp}} \to \mathscr{N}_\sigma, \qquad \boldsymbol{j}_\sigma^{\mathrm{ppsp}} : \mathcal{B}_\sigma^{\mathrm{ppsp}} \to \mathscr{J}_\sigma,$$

invariant under the corresponding pre-proto-subprimitive time reversals, preserved along the current proto-subprimitive characteristic curves, and constant on proto-subprimitive-shell-normal directions and on vertical pre-proto-subprimitive-fiber directions;

4. *exact-preserving pre-proto-subprimitive boundary potentials*

$$\boldsymbol{\rho}_\sigma^{\partial,\mathrm{ppsp}} : \mathcal{B}_\sigma^{\mathrm{ppsp}} \to \mathscr{R}_\sigma,$$

vanishing on the corresponding pre-proto-subprimitive unit sections, invariant under the corresponding pre-proto-subprimitive time reversals, annihilating vectors normal to the proto-subprimitive-generating shells, and constant on vertical pre-proto-subprimitive-fiber directions;

5. *fixed-endpoint pre-proto-subprimitive actions*

$$\mathscr{A}_\sigma^{\mathrm{ppsp}}$$

on curves constrained to the current root exact-constraint shells inside the current primitive-generating shells and current subprimitive-generating shells, whose stationary curves are exactly the curves singled out there by the restricted pre-proto-subprimitive Liouville / exact-constraint geometry, whose first variations vanish automatically along the characteristic, symmetry, vertical proto-subprimitive-fiber, and proto-subprimitive-shell-normal directions, and whose second variations are positive on smooth endpoint-fixed complements to those directions and to the normal exact-constraint directions singled out by δ_σ.

Definition 4 (Recorded exact-preservation target)**.** *The present note must preserve, without any modification,*

$$\mathscr{F}_{\mathrm{sel}}, \qquad \mathscr{F}_{\mathrm{deep}}, \qquad \mathscr{F}_{\mathrm{car}}[\mathfrak{S}, \mathfrak{R}, \mathfrak{F}], \qquad \mathscr{F}_{\mathrm{prim}}[K, J, F], \qquad \mathscr{F}_{\mathrm{mic}}[H, Y], \tag{5}$$

together with the fixed free-sector verdict

$$\Pi_{\mathrm{free}} \widehat{\Sigma}^{\mathrm{cmp}} = 0, \tag{6}$$

and hence the same previously recorded $\Pi_{\mathrm{free}} H = 0$ outcome, the same completion solve

$$\Sigma^{\mathrm{cmp}} = \Sigma^{\mathrm{cmp}}, \tag{7}$$

the same one operative metric $g^\sharp$, and the same recorded Phase D / E infrared-spectrum and universal-coupling verdicts together with the same Phase F controlled GR limit.

3. Stationary-Selection Origin of the Recorded Pre-Proto-Subprimitive Geometry

Definition 5 (Deeper stationary-selection bundles and lifted geometry)**.** *For each sector label $\sigma \in \mathsf{S}$, let*

$$\mathcal{M}_\sigma^{\mathrm{sel}}$$

be a finite-dimensional C^2 manifold of yet more primitive exact-preserving substrate states. Assume:

1. *there are smooth visible selection readout maps*

$$\Pi_\sigma^{\mathrm{sel}} : \mathcal{M}_\sigma^{\mathrm{sel}} \to \mathcal{O}_\sigma,$$

together with exact-preserving selection unit sections

$$u_\sigma^{\mathrm{sel}} : \mathcal{O}_\sigma \to \mathcal{M}_\sigma^{\mathrm{sel}}$$

satisfying

$$\Pi_\sigma^{\mathrm{sel}} \circ u_\sigma^{\mathrm{sel}} = \mathrm{id}_{\mathcal{O}_\sigma}.$$

2. *there are smooth involutive exact-preserving selection time reversals*

$$\tau_\sigma^{\mathrm{sel}} : \mathcal{M}_\sigma^{\mathrm{sel}} \to \mathcal{M}_\sigma^{\mathrm{sel}}$$

such that

$$\Pi_\sigma^{\mathrm{sel}} \circ \tau_\sigma^{\mathrm{sel}} = \Pi_\sigma^{\mathrm{sel}}, \qquad \tau_\sigma^{\mathrm{sel}} \circ u_\sigma^{\mathrm{sel}} = u_\sigma^{\mathrm{sel}}.$$

3. *each selection space carries a smooth exact-preserving positive-definite metric*

$$S_\sigma^{\mathrm{sel}}$$

preserved by the corresponding selection time reversal, and each visible selection fiber $(\Pi_\sigma^{\mathrm{sel}})^{-1}(x)$ *is compact.*

4. *there are finite-dimensional real stationary-selection target spaces*

$$\mathcal{R}_\sigma^{\mathrm{sel}}$$

with positive-definite metrics

$$r_\sigma^{\mathrm{sel}},$$

and exact-preserving smooth stationary-selection sections

$$\mathfrak{M}_\sigma^{\mathrm{sel}} : \mathcal{M}_\sigma^{\mathrm{sel}} \to \mathcal{R}_\sigma^{\mathrm{sel}}$$

preserved by the corresponding selection time reversals and vanishing on the corresponding selection unit sections. Define the stationary-selection functionals

$$\mathscr{S}_\sigma^{\mathrm{sel}}(y) \equiv \frac{1}{2}\, r_\sigma^{\mathrm{sel}}(\mathfrak{M}_\sigma^{\mathrm{sel}}(y), \mathfrak{M}_\sigma^{\mathrm{sel}}(y)). \tag{8}$$

Define the recorded pre-proto-subprimitive substrate bundles as the exact zero shells

$$\mathcal{B}_\sigma^{\mathrm{ppsp}} \equiv \left(\mathfrak{M}_\sigma^{\mathrm{sel}}\right)^{-1}(0), \tag{9}$$

and let

$$\iota_\sigma^{\mathrm{ppsp}} : \mathcal{B}_\sigma^{\mathrm{ppsp}} \hookrightarrow \mathcal{M}_\sigma^{\mathrm{sel}} \tag{10}$$

be the shell inclusion. Assume this shell is a closed embedded C^2 *submanifold, invariant under the corresponding selection time reversal, containing the image of the corresponding selection unit section, and equal to the exact set of global minima of* $\mathscr{S}_\sigma^{\mathrm{sel}}$*. Assume further that, for every* $x \in \mathcal{B}_\sigma^{\mathrm{ppsp}}$*,*

$$r_\sigma^{\mathrm{sel}}\Big(D\mathfrak{M}_\sigma^{\mathrm{sel}}(x)\xi, D\mathfrak{M}_\sigma^{\mathrm{sel}}(x)\xi\Big) \geq \zeta_\sigma^{\mathrm{sel}}\, S_\sigma^{\mathrm{sel}}(\xi,\xi), \qquad \zeta_\sigma^{\mathrm{sel}} > 0, \tag{11}$$

for all ξ *orthogonal, with respect to* S_σ^{sel}*, to* $T_x\mathcal{B}_\sigma^{\mathrm{ppsp}}$*.*

5. *there are exact-preserving lifted proto-subprimitive-generating shell maps*

$$\overline{\mathfrak{U}}_\sigma^{\mathrm{sel}} : \mathcal{M}_\sigma^{\mathrm{sel}} \to \mathcal{X}_\sigma^{\mathrm{ppsp}}$$

preserved by the corresponding selection time reversals and vanishing on the corresponding selection unit sections. Define the recorded proto-subprimitive-generating shell sections by

restriction:

$$\mathfrak{U}_\sigma^{\mathrm{ppsp}} \equiv \overline{\mathfrak{U}}_\sigma^{\mathrm{sel}}|_{\mathcal{B}_\sigma^{\mathrm{ppsp}}}. \tag{12}$$

Assume these restricted sections satisfy the same recorded positive pre-proto-subprimitive gap bounds $\omega_\sigma^{\mathrm{ppsp}} > 0$ *on the current proto-subprimitive shells cut out inside* $\mathcal{B}_\sigma^{\mathrm{ppsp}}$.

6. *there are exact-preserving lifted subprimitive-generating shell maps*

$$\overline{\mathfrak{T}}_\sigma^{\mathrm{sel}} : \mathcal{M}_\sigma^{\mathrm{sel}} \to \mathcal{T}_\sigma^{\mathrm{psp}}$$

preserved by the corresponding selection time reversals and vanishing on the corresponding selection unit sections. Define the recorded lifted subprimitive-generating shell maps by restriction:

$$\widetilde{\mathfrak{T}}_\sigma^{\mathrm{psp}} \equiv \overline{\mathfrak{T}}_\sigma^{\mathrm{sel}}|_{\mathcal{B}_\sigma^{\mathrm{ppsp}}}. \tag{13}$$

Assume these restricted maps satisfy the same recorded positive proto-subprimitive gap bounds $\nu_\sigma^{\mathrm{psp}} > 0$ *on the current subprimitive shells cut out inside* $\mathcal{B}_\sigma^{\mathrm{ppsp}}$.

7. *there are exact-preserving lifted primitive-generating shell maps*

$$\overline{\mathfrak{G}}_\sigma^{\mathrm{sel}} : \mathcal{M}_\sigma^{\mathrm{sel}} \to \mathcal{V}_\sigma^{\mathrm{subprim}}$$

preserved by the corresponding selection time reversals and vanishing on the corresponding selection unit sections. Define the recorded lifted primitive-generating shell maps by restriction:

$$\widetilde{\mathfrak{G}}_\sigma^{\mathrm{subprim}} \equiv \overline{\mathfrak{G}}_\sigma^{\mathrm{sel}}|_{\mathcal{B}_\sigma^{\mathrm{ppsp}}}. \tag{14}$$

Assume these restricted maps satisfy the same recorded positive subprimitive gap bounds $\mu_\sigma^{\mathrm{subprim}} > 0$ *on the current primitive shells cut out inside* $\mathcal{B}_\sigma^{\mathrm{ppsp}}$.

8. *there are exact-preserving lifted root-generating shell maps*

$$\overline{\mathfrak{P}}_\sigma^{\mathrm{sel}} : \mathcal{M}_\sigma^{\mathrm{sel}} \to \mathcal{W}_\sigma^{\mathrm{prim}}$$

preserved by the corresponding selection time reversals and vanishing on the corresponding selection unit sections. Define the recorded lifted root-generating shell maps by restriction:

$$\widetilde{\mathfrak{P}}_\sigma^{\mathrm{prim}} \equiv \overline{\mathfrak{P}}_\sigma^{\mathrm{sel}}|_{\mathcal{B}_\sigma^{\mathrm{ppsp}}}. \tag{15}$$

Assume these restricted maps satisfy the same recorded positive primitive gap bounds $\lambda_\sigma^{\mathrm{prim}} > 0$ *on the current root shells cut out inside* $\mathcal{B}_\sigma^{\mathrm{ppsp}}$.

9. *there are exact-preserving lifted pre-ambient-generating shell maps*

$$\overline{\mathfrak{Z}}_\sigma^{\mathrm{sel}} : \mathcal{M}_\sigma^{\mathrm{sel}} \to \mathcal{Z}_\sigma^{\mathrm{root}}$$

preserved by the corresponding selection time reversals and vanishing on the corresponding selection unit sections. Define the recorded lifted pre-ambient-generating shell maps by restriction:

$$\widetilde{\mathfrak{Z}}_\sigma^{\mathrm{root}} \equiv \overline{\mathfrak{Z}}_\sigma^{\mathrm{sel}}|_{\mathcal{B}_\sigma^{\mathrm{ppsp}}}. \tag{16}$$

Assume these restricted maps satisfy the same recorded positive root gap bounds $\kappa_\sigma^{\mathrm{root}} > 0$ on the current pre-ambient shells cut out inside $\mathcal{B}_\sigma^{\mathrm{ppsp}}$.

10. *there are exact-preserving lifted ambient-generating shell maps*

$$\overline{\mathfrak{S}}_\sigma^{\mathrm{sel}} : \mathcal{M}_\sigma^{\mathrm{sel}} \to \mathcal{H}_\sigma^{\mathrm{orig}}$$

preserved by the corresponding selection time reversals and vanishing on the corresponding selection unit sections. Define the recorded lifted ambient-generating shell maps by restriction:

$$\widetilde{\mathfrak{S}}_\sigma^{\mathrm{orig}} \equiv \overline{\mathfrak{S}}_\sigma^{\mathrm{sel}}|_{\mathcal{B}_\sigma^{\mathrm{ppsp}}}. \tag{17}$$

Assume these restricted maps satisfy the same recorded positive ambient-generating gap bounds $\epsilon_\sigma^{\mathrm{orig}} > 0$ on the current ambient shells cut out inside $\mathcal{B}_\sigma^{\mathrm{ppsp}}$.

11. *there are exact-preserving lifted shell maps*

$$\overline{\mathfrak{D}}_\sigma^{\mathrm{sel}} : \mathcal{M}_\sigma^{\mathrm{sel}} \to \mathcal{E}_\sigma$$

preserved by the corresponding selection time reversals and vanishing on the corresponding selection unit sections. Define the recorded lifted shell maps by restriction:

$$\widetilde{\mathfrak{D}}_\sigma^{\mathrm{root}} \equiv \overline{\mathfrak{D}}_\sigma^{\mathrm{sel}}|_{\mathcal{B}_\sigma^{\mathrm{ppsp}}}. \tag{18}$$

Assume these restricted maps satisfy the same recorded positive shell-gap bounds $\eta_\sigma > 0$ on the current seed shells cut out inside $\mathcal{B}_\sigma^{\mathrm{ppsp}}$.

12. *there are exact-preserving lifted exact-constraint maps*

$$\overline{\mathfrak{C}}_\sigma^{\mathrm{sel}} : \mathcal{M}_\sigma^{\mathrm{sel}} \to \mathcal{K}_\sigma$$

preserved by the corresponding selection time reversals and vanishing on the corresponding selection unit sections. Define the recorded lifted exact-constraint maps by restriction:

$$\widetilde{\mathfrak{C}}_\sigma^{\mathrm{root}} \equiv \overline{\mathfrak{C}}_\sigma^{\mathrm{sel}}|_{\mathcal{B}_\sigma^{\mathrm{ppsp}}}. \tag{19}$$

13. *there are exact-preserving cotangent realizations*

$$\Upsilon_\sigma^{\mathrm{sel}} : \mathcal{M}_\sigma^{\mathrm{sel}} \to T^*\mathcal{Q}_\sigma$$

into the same recorded configuration spaces. Let θ_{can} denote the canonical Liouville one-form on the relevant cotangent bundle. Define selection Liouville one-forms and closed two-forms by

$$\mathbf{\Lambda}_\sigma^{\mathrm{sel}} \equiv (\Upsilon_\sigma^{\mathrm{sel}})^*\theta_{\mathrm{can}}, \qquad \mathbf{\Xi}_\sigma^{\mathrm{sel}} \equiv d\mathbf{\Lambda}_\sigma^{\mathrm{sel}}. \tag{20}$$

Assume

$$(\tau_\sigma^{\mathrm{sel}})^*\mathbf{\Lambda}_\sigma^{\mathrm{sel}} = -\mathbf{\Lambda}_\sigma^{\mathrm{sel}}.$$

Define the recorded pre-proto-subprimitive Liouville one-forms and closed two-forms by shell

restriction:

$$\mathbf{\Lambda}_\sigma^{\mathrm{ppsp}} \equiv (\iota_\sigma^{\mathrm{ppsp}})^* \mathbf{\Lambda}_\sigma^{\mathrm{sel}}, \qquad \mathbf{\Xi}_\sigma^{\mathrm{ppsp}} \equiv (\iota_\sigma^{\mathrm{ppsp}})^* \mathbf{\Xi}_\sigma^{\mathrm{sel}}. \tag{21}$$

14. *there are smooth skew-Hermitian selection connection one-forms*

$$\mathcal{G}_\sigma^{\mathrm{conn}} \in \Omega^1(\mathcal{M}_\sigma^{\mathrm{sel}}; \mathfrak{u}(\mathscr{U}_\sigma))$$

that are exact-preserving on the fixed branch and satisfy the time-reversal equivariance rule

$$(\tau_\sigma^{\mathrm{sel}})^* \mathcal{G}_\sigma^{\mathrm{conn}} = -\mathcal{G}_\sigma^{\mathrm{conn}}.$$

Let

$$\mathbb{Y}_\sigma(\Gamma) \in \mathcal{U}(\mathscr{U}_\sigma)$$

denote the parallel transport generated by that selection connection one-form along any piecewise C^1 selection curve Γ. Define the recorded pre-proto-subprimitive connection one-forms and induced parallel transports by

$$\mathcal{F}_\sigma^{\mathrm{conn}} \equiv (\iota_\sigma^{\mathrm{ppsp}})^* \mathcal{G}_\sigma^{\mathrm{conn}}, \tag{22}$$

$$\mathbb{X}_\sigma(\gamma) \equiv \mathbb{Y}_\sigma(\iota_\sigma^{\mathrm{ppsp}} \circ \gamma) \tag{23}$$

on piecewise C^1 pre-proto-subprimitive curves γ.

Define the recorded pre-proto-subprimitive readout maps, pre-proto-subprimitive unit sections, pre-proto-subprimitive time reversals, and pre-proto-subprimitive metrics by restriction:

$$\Pi_\sigma^{\mathrm{ppsp}} \equiv \Pi_\sigma^{\mathrm{sel}} \circ \iota_\sigma^{\mathrm{ppsp}}, \qquad u_\sigma^{\mathrm{ppsp}} \equiv u_\sigma^{\mathrm{sel}}, \qquad \chi_\sigma^{\mathrm{ppsp}} \equiv \tau_\sigma^{\mathrm{sel}}|_{\mathcal{B}_\sigma^{\mathrm{ppsp}}}, \tag{24}$$

$$B_\sigma^{\mathrm{ppsp}} \equiv (\iota_\sigma^{\mathrm{ppsp}})^* S_\sigma^{\mathrm{sel}}. \tag{25}$$

For each visible point, define the visible pre-proto-subprimitive fibers by

$$(\Pi_\sigma^{\mathrm{ppsp}})^{-1}(x) = \left(\mathfrak{M}_\sigma^{\mathrm{sel}}\right)^{-1}(0) \cap (\Pi_\sigma^{\mathrm{sel}})^{-1}(x). \tag{26}$$

Proposition 1 (Pre-proto-subprimitive substrate bundles, shell sections, lifted maps, cotangent geometry, and connection-transports are forced)**.** *The stationary-selection data of Definition 5 recover exactly the geometric part of the recorded pre-proto-subprimitive layer.*

1. *The exact zero shells* (9) *are exactly the recorded pre-proto-subprimitive substrate bundles $\mathcal{B}_\sigma^{\mathrm{ppsp}}$, and they are exactly the global-minimum stationary loci of the selection functionals $\mathscr{S}_\sigma^{\mathrm{sel}}$.*

2. *The restricted maps* (24)–(25) *are exactly the recorded pre-proto-subprimitive readout maps, pre-proto-subprimitive unit sections, pre-proto-subprimitive time reversals, and preserved positive pre-proto-subprimitive metrics, and the visible pre-proto-subprimitive fibers* (26) *are compact.*

3. *The restricted maps* (12) *are exactly the recorded proto-subprimitive-generating shell sections, and their pre-proto-subprimitive gap bounds are the same recorded bounds $\omega_\sigma^{\mathrm{ppsp}}$ on the current proto-subprimitive shells.*

4. *The restricted maps* (13) *are exactly the recorded lifted subprimitive-generating shell maps, and their proto-subprimitive gap bounds are the same recorded bounds* $\nu_\sigma^{\mathrm{psp}}$ *on the current subprimitive shells.*

5. *The restricted maps* (14) *are exactly the recorded lifted primitive-generating shell maps, and their subprimitive gap bounds are the same recorded bounds* $\mu_\sigma^{\mathrm{subprim}}$ *on the current primitive shells.*

6. *The restricted maps* (15) *and* (16) *are exactly the recorded lifted root-generating shell maps and lifted pre-ambient-generating shell maps, with the same recorded primitive and root gap bounds.*

7. *The restricted maps* (17), (18), *and* (19) *are exactly the recorded lifted ambient-generating shell maps, lifted shell maps, and lifted exact-constraint maps with the same target metrics and the same recorded ambient, shell, and exact-constraint gap data.*

8. *The restricted forms* (21) *are exactly the recorded pre-proto-subprimitive Liouville one-forms and closed two-form geometry.*

9. *The restricted connection one-forms and parallel-transport formulas* (22)–(23) *are exactly the recorded pre-proto-subprimitive connection one-forms and their induced pre-proto-subprimitive parallel transports.*

10. *Consequently the pre-proto-subprimitive substrate bundles, their proto-subprimitive-generating shell sections, their lifted subprimitive-generating / primitive-generating / root-generating / pre-ambient-generating / ambient-generating / shell / exact-constraint maps, their pre-proto-subprimitive Liouville / closed-two-form geometry, and their pre-proto-subprimitive connection one-forms with induced parallel transports are no longer primitive at present scope; they are forced by a deeper stationary-selection / lifted-geometry / cotangent-connection layer.*

Proof. Item (1) is the stationary-selection construction itself. By assumption, the zero shell of $\mathfrak{M}_\sigma^{\mathrm{sel}}$ is a closed embedded C^2 submanifold, invariant under the corresponding selection time reversal and containing the image of the corresponding unit section. The norm-square functional (8) has this shell as its exact zero-energy locus, and the positive normal bound (11) is precisely the current-scope statement that the selected shell is not arbitrary but is the stable stationary zero set of the deeper selection principle. That is exactly the justification burden for the existence of the recorded pre-proto-subprimitive substrate bundle.

Item (2) follows by direct restriction along the shell inclusion $\iota_\sigma^{\mathrm{ppsp}}$. The restriction of a positive-definite metric to an embedded submanifold is positive definite, and compactness of the visible pre-proto-subprimitive fibers follows because they are closed subsets of compact visible selection fibers.

Items (3)–(7) are immediate from the defining restriction formulas. The target spaces and target metrics are already fixed to be the same recorded ones, and the recorded gap bounds are assumed to survive unchanged on the appropriate descendant shells. Thus no visible slot of the recorded pre-proto-subprimitive shell and lifted-map package is altered.

Item (8) follows from functoriality of pullback and exterior differentiation:

$$\boldsymbol{\Xi}_\sigma^{\mathrm{ppsp}} = (\iota_\sigma^{\mathrm{ppsp}})^* d\boldsymbol{\Lambda}_\sigma^{\mathrm{sel}} = d(\iota_\sigma^{\mathrm{ppsp}})^* \boldsymbol{\Lambda}_\sigma^{\mathrm{sel}}.$$

Hence the recorded pre-proto-subprimitive Liouville one-forms and closed two-forms are exactly the shell restrictions of the deeper selection cotangent-model Liouville data.

Item (9) follows because pullback of the skew-Hermitian connection one-form along the shell inclusion gives the recorded pre-proto-subprimitive connection one-form, while parallel transport along a curve in the shell is exactly parallel transport of its included image in the deeper selection bundle. Identity on constant curves, multiplicativity under concatenation, and inversion under time reversal are inherited without change.

Item (10) is then immediate. Every visible geometric ingredient of the recorded pre-proto-subprimitive layer is recovered verbatim, but it is no longer primitive. □

4. Stationary-Selection Basic Source and Fixed-Endpoint Origin of the Remaining Pre-Proto-Subprimitive Data

Definition 6 (Selection-basic source and fixed-endpoint action data)**.** *Assume the stationary-selection geometry of Definition 5 is fixed. Assume further, for each* $\sigma \in \mathsf{S}$*:*

1. *there are exact-preserving selection shadow one-forms*

$$\varsigma_\sigma^{\mathrm{sel}} \in \Omega^1(\mathcal{M}_\sigma^{\mathrm{sel}}; \mathcal{Y}_\sigma^{\mathrm{st}})$$

that are invariant under the corresponding selection time reversals, annihilate vectors normal to the stationary shell locus $\mathcal{B}_\sigma^{\mathrm{ppsp}}$*, annihilate vertical vectors tangent to the visible selection fibers, and are preserved along the current pre-proto-subprimitive characteristic curves singled out on the root exact-constraint shells by the restricted pre-proto-subprimitive Liouville / exact-constraint geometry. Define the recorded pre-proto-subprimitive shadow one-forms by restriction:*

$$\varsigma_\sigma^{\mathrm{ppsp}} \equiv (\iota_\sigma^{\mathrm{ppsp}})^* \varsigma_\sigma^{\mathrm{sel}}. \tag{27}$$

2. *there are exact-preserving selection rate one-forms*

$$\boldsymbol{\lambda}_\sigma^{\mathrm{sel}} \in \Omega^1(\mathcal{M}_\sigma^{\mathrm{sel}})$$

with the same invariance and annihilation properties. Define the recorded pre-proto-subprimitive rate one-forms by

$$\boldsymbol{\lambda}_\sigma^{\mathrm{ppsp}} \equiv (\iota_\sigma^{\mathrm{ppsp}})^* \boldsymbol{\lambda}_\sigma^{\mathrm{sel}}. \tag{28}$$

3. *there are exact-preserving selection density and selection flux source fields*

$$\boldsymbol{q}_\sigma^{\mathrm{sel}} : \mathcal{M}_\sigma^{\mathrm{sel}} \to \mathscr{N}_\sigma, \qquad \boldsymbol{j}_\sigma^{\mathrm{sel}} : \mathcal{M}_\sigma^{\mathrm{sel}} \to \mathscr{J}_\sigma,$$

invariant under the corresponding selection time reversals, preserved along the current pre-proto-subprimitive characteristic curves, and constant on pre-proto-subprimitive-shell-normal directions and on vertical selection-fiber directions. Define the recorded pre-proto-subprimitive density and pre-proto-subprimitive flux source fields by restriction:

$$\boldsymbol{q}_\sigma^{\mathrm{ppsp}} \equiv \boldsymbol{q}_\sigma^{\mathrm{sel}}|_{\mathcal{B}_\sigma^{\mathrm{ppsp}}}, \qquad \boldsymbol{j}_\sigma^{\mathrm{ppsp}} \equiv \boldsymbol{j}_\sigma^{\mathrm{sel}}|_{\mathcal{B}_\sigma^{\mathrm{ppsp}}}. \tag{29}$$

4. *there are exact-preserving selection boundary potentials*

$$\boldsymbol{\rho}_\sigma^{\partial,\mathrm{sel}} : \mathcal{M}_\sigma^{\mathrm{sel}} \to \mathscr{R}_\sigma$$

vanishing on the corresponding selection unit sections, invariant under the corresponding selection time reversals, annihilating vectors normal to the stationary shell locus, and constant on vertical selection-fiber directions. Define the recorded pre-proto-subprimitive boundary potentials by

$$\boldsymbol{\rho}_\sigma^{\partial,\mathrm{ppsp}} \equiv \boldsymbol{\rho}_\sigma^{\partial,\mathrm{sel}}|_{\mathcal{B}_\sigma^{\mathrm{ppsp}}}. \tag{30}$$

5. *define the selection fixed-endpoint rate actions on C^2 curves constrained to the current root exact-constraint shells inside the current primitive-generating shells, current subprimitive-generating shells, and current proto-subprimitive-generating shells by*

$$\mathscr{A}_\sigma^{\mathrm{sel}}[\Gamma] \equiv \int_\Gamma \boldsymbol{\lambda}_\sigma^{\mathrm{sel}}. \tag{31}$$

Assume their stationary curves at fixed visible endpoint data are exactly the current pre-proto-subprimitive characteristic curves determined on those shells by the restricted pre-proto-subprimitive Liouville / exact-constraint geometry, that their first variations vanish automatically along the characteristic, symmetry, vertical selection-fiber, and stationary-shell-normal directions, and that their second variations are positive on smooth endpoint-fixed complements to those directions and to the normal exact-constraint directions singled out by δ_σ. Define the recorded pre-proto-subprimitive fixed-endpoint actions by

$$\mathscr{A}_\sigma^{\mathrm{ppsp}}[\gamma] \equiv \int_\gamma \boldsymbol{\lambda}_\sigma^{\mathrm{ppsp}}. \tag{32}$$

Proposition 2 (Pre-proto-subprimitive-basic data and pre-proto-subprimitive action are forced)**.** *The selection-basic data of Definition 6 recover exactly the remaining recorded pre-proto-subprimitive layer.*

1. *The restricted one-forms* (27) *and* (28) *are exactly the recorded pre-proto-subprimitive shadow and pre-proto-subprimitive rate one-forms.*

2. *The restricted source fields* (29) *are exactly the recorded pre-proto-subprimitive density and pre-proto-subprimitive flux source fields.*

3. *The restricted boundary potentials* (30) *are exactly the recorded pre-proto-subprimitive boundary potentials.*

4. *The fixed-endpoint actions* (32) *are exactly the recorded pre-proto-subprimitive actions, and their inherited second-variation positivity is exactly the recorded fixed-endpoint pre-proto-subprimitive rate-action geometry.*

5. *Consequently the pre-proto-subprimitive-basic shadow / rate / density / flux / boundary data and the fixed-endpoint pre-proto-subprimitive rate-action geometry are no longer primitive at present scope; they are forced by deeper selection-basic restriction and the same fixed-endpoint stationary-selection variational geometry.*

Proof. Item (1) follows from the restriction construction itself. Because the selection shadow and selection rate one-forms annihilate both the stationary-shell-normal directions and the vertical directions tangent to the visible selection fibers, their restrictions to the pre-proto-subprimitive zero shells are well defined and already satisfy the invariance and horizontality required at pre-proto-subprimitive level.

Item (2) is immediate from the restriction formula (29). Constancy on pre-proto-subprimitive-shell-normal directions and on vertical selection-fiber directions guarantees that the restricted source fields depend only on the pre-proto-subprimitive states themselves, while the assumed characteristic-flow and time-reversal invariance give exactly the recorded exact-preserving behavior.

Item (3) is the same argument applied to the boundary potentials. Since the selection boundary potentials vanish on the selection unit sections, their pre-proto-subprimitive restrictions vanish on the recorded pre-proto-subprimitive unit sections as required.

Item (4) follows from fixed-endpoint stationary reduction. The selection rate actions are assumed to have precisely the current pre-proto-subprimitive characteristic curves as stationary curves, with first variations vanishing automatically along the characteristic, symmetry, vertical selection-fiber, and stationary-shell-normal directions. Restricting those actions to curves in the stationary zero shells yields exactly the recorded pre-proto-subprimitive-action formula (32). The positive second variation on endpoint-fixed complements descends directly to the recorded pre-proto-subprimitive stationary geometry.

Item (5) is therefore immediate. None of the current pre-proto-subprimitive-basic data or pre-proto-subprimitive fixed-endpoint stationary geometry needs to be inserted separately once the selection-basic layer is fixed. □

5. Exact Recovery of the Recorded Pre-Proto-Subprimitive Layer and Downstream Package

Theorem 1 (Same pre-proto-subprimitive input layer)**.** *Under the fixed branch of Definition 1, the deeper stationary-selection / cotangent-connection / source-action construction recovers exactly the pre-proto-subprimitive input layer of Definitions 2 and 3.*

Proof. The geometric part is Proposition 1. The remaining basic-data and fixed-endpoint-action part is Proposition 2. Together they recover every recorded item without alteration. □

Theorem 2 (Same proto-subprimitive layer and same downstream package)**.** *Under the fixed branch of Definition 1, the deeper stationary-selection / cotangent-connection / source-action construction preserves without change:*

1. *the same pre-proto-subprimitive stationary-selection / lifted connection-cotangent / source-action input layer;*
2. *the same proto-subprimitive substrate / subprimitive-generating shell / lifted connection-cotangent / parallel-transport input layer;*
3. *the same subprimitive substrate / primitive-generating shell / lifted connection-cotangent / parallel-transport input layer;*
4. *the same primitive substrate / root-generating shell / lifted connection-cotangent input layer;*

5. *the same root-shell / lifted connection-cotangent / parallel-transport input layer;*
6. *the same pre-ambient-shell / connection-cotangent / parallel-transport input layer;*
7. *the same ambient-shell / cotangent-Liouville / connection-holonomy input layer;*
8. *the same constrained substrate-bundle / exact-constraint / Liouville-holonomy input layer;*
9. *the same proto-process / stationary-selection / characteristic-bundle input layer;*
10. *the same process-state / reversible-line-field / compact-calibration-fiber / basic-form layer;*
11. *the same reversible-process-thread / calibration-generator / shadow-current / rate-density / boundary-recorder layer;*
12. *the same reversible-process-window / calibration-action / stationary-shadow / local-rate / endpoint-recorder layer;*
13. *the same reversible-history / stationary-equivalence / short-history-action / endpoint-readout / cotangent-lift layer;*
14. *the same reversible-groupoid / transport-action / constrained-stationary package;*
15. *the same reconfiguration-group / transport-law / equilibrium-reduction package;*
16. *the same derivation-algebra / balance-law / equilibrium-manifold layer;*
17. *hence the same process-state and reversible process-thread / process-window / history / groupoid / reconfiguration / derivation chain;*
18. *the same* $\mathscr{F}_{\mathrm{sel}}$*;*
19. *the same exact-preservation normal form* $\mathscr{F}_{\mathrm{deep}}$*;*
20. *the same carrier-origin functional* $\mathscr{F}_{\mathrm{car}}[\mathfrak{S}, \mathfrak{R}, \mathfrak{F}]$*;*
21. *the same primitive reservoir* $\mathscr{F}_{\mathrm{prim}}[K, J, F]$*;*
22. *the same microscopic-equilibrium reservoir* $\mathscr{F}_{\mathrm{mic}}[H, Y]$*;*
23. *the same free-sector verdict* $\Pi_{\mathrm{free}}\widehat{\Sigma}^{\mathrm{cmp}} = 0$*, and hence the same* $\Pi_{\mathrm{free}}H = 0$ *outcome;*
24. *the same substrate and observer completion solves* (4)*;*
25. *the same* Σ^{cmp} *solve, the same observer-equation closure, the same one operative metric* $g^{\sharp}$*, and the same recorded Phase D / E and Phase F gains.*

Proof. Theorem 1 reproduces exactly the input layer required by the immediately preceding proto-subprimitive-substrate / subprimitive-generating shell / lifted connection-cotangent / parallel-transport origin note. That note already proves that once this pre-proto-subprimitive input layer is fixed, the same proto-subprimitive layer follows without change, and with it the same full downstream chain. Since none of the visible slots, elimination steps, or completion equations are altered here, every downstream object listed above is reproduced verbatim. □

Corollary 1 (Recorded gain package is preserved)**.** *Because the same downstream package is recovered, the recorded Phase D / E infrared-spectrum and universal-coupling verdicts, the recorded Phase F controlled GR limit, and the already recorded observer-equation closure package remain preserved without modification.*

Proof. This is the direct content of Theorem 2. □

6. Claim-Boundary Verdict

Theorem 3 (Pre-proto-subprimitive stationary-selection origin verdict)**.** *At the disciplined scope of the present note, the exact positive result is the following.*

1. *The exact-preserving pre-proto-subprimitive substrate bundles* $\mathcal{B}^{\mathrm{ppsp}}_{\mathrm{car}}, \mathcal{B}^{\mathrm{ppsp}}_{\mathrm{cur}}, \mathcal{B}^{\mathrm{ppsp}}_{\mathrm{hom}}$ *are no longer primitive at current scope.*

2. *They arise as the exact stationary zero shells of deeper selection sections* $\mathfrak{M}^{\mathrm{sel}}_{\mathrm{car}}, \mathfrak{M}^{\mathrm{sel}}_{\mathrm{cur}}, \mathfrak{M}^{\mathrm{sel}}_{\mathrm{hom}}$ *on compact visible selection bundles, equivalently as the zero-energy stationary loci of the deeper selection functionals* $\mathscr{S}^{\mathrm{sel}}_{\mathrm{car}}, \mathscr{S}^{\mathrm{sel}}_{\mathrm{cur}}, \mathscr{S}^{\mathrm{sel}}_{\mathrm{hom}}$.

3. *The visible pre-proto-subprimitive readout maps, pre-proto-subprimitive unit sections, pre-proto-subprimitive time reversals, and preserved positive pre-proto-subprimitive metrics are no longer primitive either.*

4. *They are the exact-shell restrictions of deeper selection readouts, units, time reversals, and selection metrics, and the recorded compact visible pre-proto-subprimitive fibers are closed stationary-shell slices inside compact visible selection fibers.*

5. *The proto-subprimitive-generating shell sections are no longer primitive either.*

6. *They arise as the pre-proto-subprimitive-shell restrictions of deeper lifted proto-subprimitive-generating shell maps into the same positive target spaces with the same target metrics* $x^{\mathrm{ppsp}}_{\mathrm{car}}, x^{\mathrm{ppsp}}_{\mathrm{cur}}, x^{\mathrm{ppsp}}_{\mathrm{hom}}$.

7. *The lifted subprimitive-generating shell maps, lifted primitive-generating shell maps, lifted root-generating shell maps, lifted pre-ambient-generating shell maps, lifted ambient-generating shell maps, lifted shell maps, and lifted exact-constraint maps are no longer primitive either.*

8. *They arise as the pre-proto-subprimitive-shell restrictions of deeper lifted maps into the same recorded target spaces with the same target metrics.*

9. *The pre-proto-subprimitive cotangent-model Liouville one-forms* $\mathbf{\Lambda}^{\mathrm{ppsp}}_{\mathrm{car}}, \mathbf{\Lambda}^{\mathrm{ppsp}}_{\mathrm{cur}}, \mathbf{\Lambda}^{\mathrm{ppsp}}_{\mathrm{hom}}$ *and the pre-proto-subprimitive closed two-form geometry* $\mathbf{\Xi}^{\mathrm{ppsp}}_{\mathrm{car}}, \mathbf{\Xi}^{\mathrm{ppsp}}_{\mathrm{cur}}, \mathbf{\Xi}^{\mathrm{ppsp}}_{\mathrm{hom}}$ *are no longer primitive either.*

10. *They arise as the pre-proto-subprimitive-shell restrictions of deeper selection cotangent-model Liouville one-forms* $\mathbf{\Lambda}^{\mathrm{sel}}_{\mathrm{car}}, \mathbf{\Lambda}^{\mathrm{sel}}_{\mathrm{cur}}, \mathbf{\Lambda}^{\mathrm{sel}}_{\mathrm{hom}}$ *and their exact two-forms on the selection bundles.*

11. *The pre-proto-subprimitive connection one-forms* $\mathcal{F}^{\mathrm{conn}}_{\mathrm{car}}, \mathcal{F}^{\mathrm{conn}}_{\mathrm{cur}}, \mathcal{F}^{\mathrm{conn}}_{\mathrm{hom}}$ *and their induced parallel transports* $\mathbb{X}_{\mathrm{car}}, \mathbb{X}_{\mathrm{cur}}, \mathbb{X}_{\mathrm{hom}}$ *are no longer primitive either.*

12. *They arise as pre-proto-subprimitive-shell restrictions of deeper skew-Hermitian selection connection one-forms* $\mathcal{G}^{\mathrm{conn}}_{\mathrm{car}}, \mathcal{G}^{\mathrm{conn}}_{\mathrm{cur}}, \mathcal{G}^{\mathrm{conn}}_{\mathrm{hom}}$ *and of their induced selection parallel transports.*

13. *The pre-proto-subprimitive-basic shadow / rate / density / flux / boundary data and the fixed-endpoint pre-proto-subprimitive rate-action geometry are no longer primitive either.*

14. *They arise as pre-proto-subprimitive-shell restrictions of deeper selection-basic source data and the same fixed-endpoint stationary-selection variational geometry.*

15. *Therefore the full pre-proto-subprimitive layer of the immediately preceding proto-subprimitive note is recovered without change.*

16. *Consequently the same proto-subprimitive layer, the same subprimitive layer, the same primitive layer, the same root layer, the same pre-ambient layer, the same ambient-shell / cotangent-Liouville / connection-holonomy layer, the same process-state and reversible-chain package, the same* $\mathscr{F}_{\mathrm{sel}}$*, the same exact-preservation normal form, the same carrier-origin / primitive-reservoir / microscopic-equilibrium functionals, the same* Σ^{cmp} *solve, the same* $\Pi_{\mathrm{free}} H = 0$ *outcome, the same observer-equation closure, the same one operative metric, and the same recorded Phase D / E and Phase F gains remain unchanged.*

17. *Stronger promotion language is still not justified here, because the present note explains only why the previous pre-proto-subprimitive layer arises from a deeper stationary-selection / cotangent-connection / source-action layer; it does not yet derive why that new deeper selection layer is itself unique, final, or inevitable beyond current scope.*

18. *No broader packaging consequence or default side-line continuation follows from the mathematical content of this note alone. The remaining honest burden is one layer deeper again on this same exact-preservation line, or else another comparably concrete observer-equation gain on the same one-metric route.*

Proof. Items (1)–(12) are Proposition 1. Items (13) and (14) are Proposition 2. Items (15) and (16) are Theorems 1 and 2 together with Corollary 1. Items (17) and (18) are the present claim-boundary discipline stated in the abstract and introduction. □

Corollary 2 (What remains next)**.** *The primary active burden moves one layer deeper again. It is no longer to justify why the pre-proto-subprimitive substrate bundles, their proto-subprimitive-generating shell sections, their lifted subprimitive-generating / primitive-generating / root-generating / pre-ambient-generating / ambient-generating / shell / exact-constraint maps, their pre-proto-subprimitive Liouville / closed-two-form geometry, their connection one-forms and induced parallel transports, or their pre-proto-subprimitive-basic shadow / rate / density / flux / boundary data and fixed-endpoint pre-proto-subprimitive rate-action geometry are admissible at current scope; that step is now recorded. The next honest burden is either to derive or justify why the deeper stationary-selection bundles, their exact stationary-selection sections and functionals, their lifted proto-subprimitive-generating / subprimitive-generating / primitive-generating / root-generating / pre-ambient-generating / ambient-generating / shell / exact-constraint maps, their selection cotangent-model Liouville one-forms, their selection connection one-forms and parallel transports, and their selection-basic source / action data are themselves forced by yet more primitive substrate structure, or else to produce another comparably concrete observer-equation gain on the same one-metric line. No stronger promotion language, no*

packaging pass, and no default move onto the anchored positive-pair side line are justified unless they directly discharge one of those burdens.

7. Conclusion

The positive result of this note is that the pre-proto-subprimitive layer is no longer left primitive at present scope. It is recovered from a still deeper stationary-selection / cotangent-connection / source-action layer.

At that deeper level, the recorded pre-proto-subprimitive substrate bundles

$$\mathcal{B}^{\mathrm{ppsp}}_{\mathrm{car}}, \qquad \mathcal{B}^{\mathrm{ppsp}}_{\mathrm{cur}}, \qquad \mathcal{B}^{\mathrm{ppsp}}_{\mathrm{hom}}$$

are exact stationary zero shells inside compact visible selection bundles. Their readout maps, unit sections, time reversals, and preserved positive metrics are exact-shell restrictions of deeper selection data. Their proto-subprimitive-generating shell sections are pre-proto-subprimitive-shell restrictions of deeper lifted proto-subprimitive-generating shell maps into the same positive target spaces with the same target metrics. Their lifted subprimitive-generating shell maps, lifted primitive-generating shell maps, lifted root-generating shell maps, lifted pre-ambient-generating shell maps, lifted ambient-generating shell maps, lifted shell maps, and lifted exact-constraint maps are pre-proto-subprimitive-shell restrictions of deeper lifted maps into the same target spaces with the same target metrics. Their pre-proto-subprimitive Liouville one-forms and closed two-form geometry are pre-proto-subprimitive-shell restrictions of deeper selection cotangent-model Liouville data. Their pre-proto-subprimitive connection one-forms and induced parallel transports are pre-proto-subprimitive-shell restrictions of deeper skew-Hermitian selection connection one-forms and their selection parallel transports. Their pre-proto-subprimitive-basic shadow / rate / density / flux / boundary data and their fixed-endpoint pre-proto-subprimitive rate-action geometry are pre-proto-subprimitive-shell restrictions of deeper selection-basic data and the same fixed-endpoint stationary-selection variational geometry.

Because that deeper construction reproduces the full pre-proto-subprimitive layer without changing any of its visible slots, the immediately preceding proto-subprimitive-substrate / subprimitive-generating shell / lifted connection-cotangent / parallel-transport origin note applies verbatim. Consequently the same proto-subprimitive layer, the same subprimitive layer, the same primitive layer, the same root layer, the same pre-ambient layer, the same ambient-shell / cotangent-Liouville / connection-holonomy layer, the same process-state and reversible-chain package, the same $\mathscr{F}_{\mathrm{sel}}$, the same exact-preservation normal form, the same carrier-origin / primitive-reservoir / microscopic-equilibrium functionals, the same Σ^{cmp} solve, the same $\Pi_{\mathrm{free}} H = 0$ outcome, the same observer-equation closure, the same one operative metric, and the same recorded Phase D / E and Phase F gains remain unchanged.

The scope is still disciplined rather than final. The present note does not prove that the deeper stationary-selection / cotangent-connection / source-action layer is unique or ultimate. The next honest burden is to derive or justify why that new deeper layer itself is forced by still more primitive substrate structure, or else to record another comparably concrete observer-equation gain on the same one-metric line.